

KNN Task Example

Problem Statement:

You are given the following dataset of students and whether they passed or failed an exam based on the number of hours they studied and the number of practice tests they took.

Student	Hours Studied	Practice Tests Taken	Result
A	2	1	Fail
B	4	1	Fail
C	6	2	Pass
D	8	3	Pass
E	10	4	Pass

You need to predict the result (Pass/Fail) for a new student who studied for 5 hours and took 2 practice tests, using $K = 3$ and Euclidean distance.

Student Task:

1. Calculate the Euclidean distance between the new student and each student in the dataset.
2. Pick the 3 nearest neighbors ($K = 3$).
3. Based on the majority result (Pass or Fail), predict the result for the new student.
4. Show all steps and final prediction.